

```
from google.colab import files
uploaded = files.upload()
```

Screenshot... 142745.png

Screenshot 2026-08-07 142745.png(image/png) - 94094 bytes, last modified: 8/7/2026 - 100% done
Saving Screenshot 2026-08-07 142745.png to Screenshot 2026-08-07 142745.png

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

# Read the image
# The 'image.jpg' file does not exist in the Colab environment.
# Instead, use the name of the file that was uploaded via `files.upload()`.
# The variable 'uploaded_filename' holds this name.
img = cv2.imread(uploaded_filename)

# Check if image was loaded successfully
if img is None:
    print(f"Error: Could not load image '{uploaded_filename}'. Please ensure it's a valid image.")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Laplacian mask with diagonal neighbors
    laplacian_mask = np.array([
        [1, 1, 1],
        [1, -8, 1],
        [1, 1, 1]
    ], dtype=np.float32)

    # Apply Laplacian filter
    laplacian = cv2.filter2D(gray, cv2.CV_32F, laplacian_mask)

    # Sharpen the image
```

```
sharpened = gray.astype(np.float32) - laplacian
```

```
# Convert to 8-bit
```

```
sharpened = cv2.convertScaleAbs(sharpened)
```

```
# Display images
```

```
plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
```

```
plt.imshow(gray, cmap="gray")
```

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
```

```
plt.imshow(sharpened, cmap="gray")
```

```
plt.title("Sharpened Image")
```

```
plt.axis("off")
```

```
plt.show()
```

Original Image



Sharpened Image



